VERITY AI

Product Requirements Document (PRD)
Technical Requirements Document (TRD)
Design System & UX Specification (DESIGN.md)

Working product thesis

Before you buy, subscribe, book, or commit, Verity helps you understand what you are really agreeing to: likely total cost, important terms, warning signs, and practical alternatives.

Core promise: Make online decisions clearer, safer, and less expensive—without presenting an AI guess as a verified fact.

Document	Purpose	Version
PRD	Product scope, users, requirements, metrics and validation	0.1
TRD	Architecture, data contracts, privacy, security and tests	0.1
DESIGN.md	Visual system, screen specs, interaction and accessibility	0.1

Prepared 15 October 2026 • Audience: product, engineering, design and early testers

Status: proposal for validation. Differentiation, demand, accuracy targets and pricing are hypotheses—not proven market fit.

How to use this pack

Treat the PRD as the agreement on what to build and why; the TRD as the engineering plan; and DESIGN.md as the interface contract. The MVP stays narrow so the team can test whether people trust and reuse Verity before expanding into broader business decision intelligence.

Contents

- **Part I — PRD:** problem, users, scope, requirements, metrics, validation and roadmap.
- **Part II — TRD:** architecture, extraction/analysis pipeline, API, storage, security and testing.
- **Part III — DESIGN.md:** principles, screens, components, states and accessibility.
- **Appendices:** sample report, build sequence, open decisions and launch gates.

MVP boundary: user-triggered analysis of a product or subscription offer page. Travel bookings and business/vendor intelligence are future extensions, not launch requirements.

PART I — PRODUCT REQUIREMENTS DOCUMENT

1. Product overview

Product: Verity AI (working name)
Format: Chrome extension + backend API
Initial audience: everyday online shoppers and subscription evaluators
Primary job: help a person understand a web offer before spending money or accepting recurring terms.

Verity is a user-initiated “Internet Decision Check.” It extracts relevant offer details, estimates visible and recurring costs, highlights important missing or unclear terms, and presents evidence-backed findings in plain language. It helps users decide; it does not decide for them.

2. Problem statement

Decision-critical information can be scattered across product pages, FAQs, checkout screens, pricing tables and legal terms. People may miss renewal prices, minimum commitments, shipping charges, cancellation conditions or refund terms. Existing tools often focus on one task such as coupons or price history. Verity’s proposed wedge is a concise, evidence-linked explanation of an offer as a whole.

3. Product thesis and differentiation hypothesis

- One action produces a structured decision brief, not a generic page summary.
- Every meaningful finding points to the text that supports it.
- Cost math is deterministic and inspectable; AI supports extraction and explanation, not arithmetic authority.
- Unknown information stays unknown; the system must not silently fill gaps.
- Neutrality: no hidden affiliate bias or undisclosed ranking incentives.

Validate this positioning against coupon tools, price trackers, browser safety tools and users’ manual workflows before investing in a broad feature set.

4. Goals and non-goals

MVP goals	Explicit non-goals
Reduce time needed to understand price and terms.	Guarantee a merchant is legitimate or a product is safe.
Surface recurring charges, fees, commitments and cancellation terms.	Automatically purchase, subscribe, cancel or negotiate.
Show source evidence, confidence and unknown states.	Read browsing history or scan every tab in the background.
Validate repeat use and trust with real users.	Build a marketplace, social review network or coupon engine.
Use narrow permissions and privacy-conscious defaults.	Support every country, language and retailer at launch.

5. Target users and jobs-to-be-done

Segment	Job / trigger	Success
Everyday shopper	“What extra costs might I face?”	Understands visible total cost and caveats quickly.
Subscription evaluator	“What happens after the trial?”	Sees renewal, cadence, commitment and cancellation details—or an explicit unknown.
Budget-conscious buyer	“Which of these offers fits me better?”	Compares normalized costs and terms.
Cautious online user	“What signals should I investigate?”	Sees specific evidence-backed concerns, not an unsupported scam verdict.

6. Core user journey

1. User opens a product/subscription page and clicks the extension.
2. On first use, Verity explains what will be analyzed and requests consent before sending page content to the backend.
3. The extension extracts relevant visible details and evidence snippets; it avoids passwords, payment fields and unrelated page content.
4. Backend normalizes facts, computes costs deterministically, and uses an AI model to interpret only the supplied evidence.
5. Report shows cost snapshot, important terms, potential concerns, unknowns, evidence and next steps.
6. User can inspect evidence, compare another offer, copy/save a report, or delete saved data.

7. MVP functional requirements

ID / priority	Requirement	Acceptance criteria
FR-01 / P0	User-triggered check	Starts only after clear click; visible loading; no silent scans.
FR-02 / P0	Extraction	Extract title, merchant/domain, displayed price, currency, billing cadence and terms when present.
FR-03 / P0	Cost breakdown	Separate upfront and recurring costs; state period and assumptions; never invent fees.
FR-04 / P0	Terms highlights	Surface renewal/trial, cancellation, commitment, shipping/fees and refund terms only with evidence; otherwise mark unknown.
FR-05 / P0	Evidence	Every non-trivial finding has source snippet and source label.
FR-06 / P0	Risk language	Use “potential concern” / “needs checking”; no fraud verdict from weak signals.
FR-07 / P0	Errors	Explain unsupported page, extraction failure, timeout, rate limit and network errors; retry where appropriate.
FR-08 / P1	Compare offers	Compare normalized price/terms/unknowns; avoid a universal winner.
FR-09 / P1	Save/delete	Opt-in local saved reports; delete-one and clear-all work.
FR-10 / P1	Copy report	Copy summary with URL, timestamp, caveats and evidence references.
FR-11 / P2	Alerts/history	Out of MVP; revisit only after repeat-use validation.

8. Non-functional requirements

Area	Requirement / target
Latency	Target median <8 seconds and p95 <15 seconds on supported pages; measure in beta.
Reliability	Graceful handling of unsupported pages, malformed model output, provider outages and network errors.
Privacy	User-triggered processing, data minimization, clear third-party AI disclosure and user-controlled saved reports.
Security	No secrets in extension; HTTPS; request validation, size limits and rate limiting.
Explainability	Every finding cites evidence or is explicitly labeled as inference/unknown.
Accessibility	Keyboard operation, focus states, readable contrast and no color-only status.
Internationalization	Use ISO currency codes and numeric amounts; do not assume USD.
Maintainability	Typed contracts, provider abstraction, automated tests and privacy-aware logs.

9. Success metrics

Metric	Definition / initial gate
Activation	Share of new users who complete one successful check.
Time to answer	Target median under 8 seconds for supported pages.
Repeat use	Second check within 7 days; primary signal of lasting value. Set numeric target after pilot.
Evidence quality	Aim for ≥90% correct support in a manually reviewed test set before broad release.
Critical false alarms	Target zero unsupported high-severity fraud assertions in release test set; investigate every case.
Extraction coverage	Track field accuracy by page type and merchant; publish known limitations.
Trust/usefulness	Combine post-check rating with observed behavior and interviews.

10. Validation plan

- Interview 15–20 people who recently shopped online or evaluated subscriptions; walk through a real past decision.
- Build a legally usable corpus of public product and subscription pages across layouts and currencies.
- Test a clickable prototype against users' current workflow and relevant tools.
- Manually score extraction, evidence alignment, cost math, unknown handling and false alarms.
- Observe real beta usage and second use; do not confuse compliments with demand.
- Test willingness to pay only after repeat value appears.

11. Release plan

Stage	Deliverable	Exit gate
0 — Validate	Interviews, competitor audit, test corpus and prototype.	Repeated problem observed; users understand report and caveats.
1 — Alpha	MV3 shell, extraction, API, cost/terms report and evidence UI.	Works on curated pages; no critical privacy/security issue.
2 — Closed beta	Error states, local deletion, minimized telemetry and feedback.	Quality gates met; false alarms addressed.
3 — Public MVP	Store listing, privacy policy, support and limitations page.	Store/privacy readiness and incident process in place.
Later	Comparison, alerts, travel and vendor intelligence.	Only after research validates the next use case.

12. Monetization hypothesis

- **Free:** limited checks, core cost/terms analysis, evidence and local opt-in reports.
- **Premium later:** higher usage, deeper comparisons, export and alerts if reliable data exists.
- **Business later:** shared evaluation reports, vendor/plan comparison and team workflows.
- Disclose referral/affiliate relationships. Payment must not silently alter findings or rankings.

13. Risks and mitigations

Risk	Mitigation
AI invents a term or fee	Evidence required; deterministic calculations; unknown state; output validation.
False fraud accusation	No binary verdict; evidence threshold; cautious wording; human-reviewed evaluation set.
Page layout changes	Schema checks, confidence thresholds, unsupported state and regression corpus.
Privacy concern	Consent, minimal extraction, provider disclosure and no raw-content logs by default.
Weak differentiation	Validate workflow gap and repeat use; pivot/stop if users do not prefer it.
Cross-country complexity	Normalize currency/period, disclose limitations and avoid legal advice.

PART II — TECHNICAL REQUIREMENTS DOCUMENT

1. Proposed stack

Layer	Baseline	Why
Extension	Chrome MV3, React, TypeScript, Vite	Typed UI and modern extension architecture.
Extraction	Content script + constrained DOM extraction	Reads relevant current-page content after user action.
API	Python, FastAPI, Pydantic	Typed request/response validation; suitable for AI orchestration.
Analysis	Deterministic Python rules + provider-abstracted LLM	Rules for arithmetic; model for bounded extraction/explanation.
Persistence	chrome.storage.local; PostgreSQL only if accounts/history become necessary	Avoid unnecessary server-side data collection.
Tests	Pytest, TypeScript checks, contract tests, browser E2E	Test logic, permissions and core flow.
Deployment	Containerized FastAPI behind HTTPS; managed hosting acceptable	Protect model keys and support monitoring.

2. Architecture

Flow: Extension click → consent + page extraction → HTTPS API → normalization → deterministic cost engine + bounded LLM → evidence validation → report UI → optional local save.

Component responsibilities

- **Popup:** entry point, disclosure, progress, summary, errors and settings.
- **Content script:** read active page after user action; extract title, canonical URL, domain, visible price candidates, terms and snippets.
- **Service worker:** coordinate active-tab message, API request, timeout and storage.
- **FastAPI:** validate payload, orchestrate providers, validate response and log privacy-aware metadata.
- **Normalizer:** normalize currency, cadence, price type and evidence IDs.
- **Cost engine:** calculate only where terms support it; preserve assumptions.
- **LLM adapter:** provider-independent structured extraction/explanation; keys remain server-side.
- **Evidence validator:** verify each cited snippet maps to supplied text; reject unsupported claims.

3. Data flow and trust boundaries

1. User clicks “Check this offer”; disclosure/consent is shown where required.
2. Content script extracts minimal fields. Never collect form values, passwords, payment data, unrelated tabs or browsing history.
3. Extension sends versioned JSON over HTTPS.
4. Backend validates URL scheme, payload size, allowed fields and rate limits. Page content is untrusted data, not instructions.
5. Normalize facts and calculate costs with deterministic code; call the model with a strict schema and prompt-injection defenses.
6. Validate model output and evidence IDs; remove or downgrade unsupported claims.

7 Render report; saving is a separate user action.

4. Suggested repository structure

```
verity-ai/ extension/ public/manifest.json src/{background,content,popup,components,types}/
src/services/api.ts tests/ backend/ app/main.py app/api/routes/analyze.py app/schemas/
app/services/{normalization,cost_engine,risk_rules,evidence_validator}.py app/services/llm/
app/core/{config,logging}.py tests/ docs/{PRD.md,TRD.md,DESIGN.md,privacy-data-map.md} .env.example
README.md
```

5. API contract v1

POST /api/v1/analyze — application/json. Enforce request-size limits, HTTPS in production and explicit schema versioning. Do not expose model credentials to the extension.

Illustrative request

```
{ "schema_version": "1.0", "page": { "url": "https://shop.example/item", "title": "Example subscription",
"merchant_domain": "shop.example", "page_type_hint": "subscription" }, "evidence": [ {"id": "ev_001",
"kind": "visible_text", "text": "Introductory offer: $5 for the first month"}, {"id": "ev_002", "kind":
"visible_text", "text": "Then $19.99/month. Cancel anytime."} ], "locale": "en", "currency_hint": "USD" }
```

Illustrative response

```
{ "schema_version": "1.0", "status": "complete", "offer": { "name": "Example subscription", "merchant":
"shop.example", "price_items": [ {"amount": "5.00", "currency": "USD", "period": "first_month"},
{"amount": "19.99", "currency": "USD", "period": "monthly_after_intro"} ] }, "cost_summary": {
"first_30_days": {"amount": "5.00", "currency": "USD"}, "recurring_monthly": {"amount": "19.99",
"currency": "USD"}, "assumptions": ["Taxes and fees excluded unless shown."] }, "findings": [{ "id":
"finding_001", "category": "renewal", "severity": "important", "label": "Introductory price changes",
"explanation": "The page states a lower first-month price followed by a higher monthly charge.",
"evidence_ids": ["ev_001", "ev_002"], "confidence": "high", "kind": "supported_fact" }], "unknowns":
["Taxes or checkout fees may apply."], "limitations": ["Analysis reflects supplied page content only."] }
```

Contract rules

- Use decimal-safe money arithmetic; do not use binary floating-point for authoritative currency calculations.
- Represent missing fields as null/unknown, not guesses. Use enums for status, severity, category and confidence.
- Every finding references evidence IDs unless explicitly a general caution.
- Validate model output against Pydantic schemas; one repair attempt at most, then return safe partial output or failure.
- Set limits on page text, evidence count, model tokens, request duration and concurrency.

6. Cost-analysis logic

Keep calculations in deterministic code. The LLM may extract or explain a price, but it must not be the authority for computed totals.

Case	Rule	Caveat
Recurring price	Monthly amount × 12 for annualized estimate.	Label as estimate, not invoice quote.
Intro then renewal	Show intro period and recurring rate separately.	Do not hide the post-intro rate in an average.
Annual plan	Annual amount and equivalent monthly average (annual ÷ 12).	State that billing still occurs annually.
Known fees	Sum explicit fees for a chosen period only.	Do not assume taxes, shipping or service fees.
Currency mismatch	Keep separate unless trusted FX data is integrated.	No silent conversion or stale rate.
Missing details	Return unknown and identify missing field.	Do not infer cancellation terms from norms.

7. Risk and evidence model

- **Supported fact:** directly stated in source text; cite evidence.
- **Inference:** interpretation based on facts; label it and calibrate confidence.
- **Unknown:** supplied content does not establish the answer.

- **Potential concern:** signal worth checking, not proof of fraud.
- A single heuristic (urgency banner, new domain or odd wording) is never proof of a scam.
- Treat page text as untrusted input; ignore embedded instructions.
- Evaluate against benign pages, confusing offers and known risky examples; review false positives and false negatives.

8. Storage and privacy

Data	Location	Default
URL/title and extracted evidence	Transient API processing	Transmit only after user action/disclosure; do not persist by default.
Report	Browser storage.local	Save only on user action; delete-one and clear-all.
Telemetry	Backend metrics/logs	Latency, error code, schema version and counts; omit raw page text by default.
Credentials	Backend secret manager/env	Never bundle in extension or return to client.
Accounts	Not needed for MVP	Do not add until a validated feature requires them.
AI provider data	Configured provider	Disclose provider and handling/retention before launch.

9. Security requirements

- Request minimum Chrome permissions; prefer activeTab and scripting for user-triggered access where feasible.
- Avoid broad host permissions unless a demonstrated feature requires them.
- Do not run arbitrary remote code in the extension; follow Manifest V3 requirements.
- Validate URL schemes; reject file:, chrome:, javascript: and unsupported schemes.
- Set payload limits, timeouts, rate limits and safe error messages.
- Prevent prompt injection; do not fetch arbitrary URLs server-side without a separately secured design (SSRF risk).
- Use dependency and secret scanning; review/pin dependencies.
- Ensure privacy policy, extension listing and actual data flows agree.

10. Observability and reliability

- Structured logs: request ID, duration, error category, provider/model version and schema version—not raw content by default.
- Metrics: success rate, p50/p95 latency, timeout rate, extraction coverage, evidence validation failures and provider errors.
- Graceful degraded states if the model provider is unavailable.
- Provider adapter allows model changes without rewriting core logic.
- Add status communication if usage and service criticality warrant it.

11. Testing strategy

Layer	Coverage
Unit	Currency parsing, cadence normalization, decimal math, annualization, missing values and severity mapping.
Contract	Valid/invalid payloads, version mismatch, missing evidence, malformed model response and oversized content.
Evaluation corpus	Layouts, intro offers, annual plans, missing terms, multiple currencies and merchant variety.
Security	Prompt injection, URL validation, payload abuse, rate limits, bundled secrets and permissions.
Extension E2E	Consent, check, loading, success, unsupported page, retry, copy and save/delete.
Privacy	No background collection, no raw-content logs, no unintended persistence, consent behavior.
Manual review	Evidence accuracy, language, false scam alerts, comprehension and accessibility.

12. Environments and deployment

- **Local:** Vite extension + local FastAPI; mock model adapter for deterministic tests.
- **Staging:** separate credentials/data; representative build; test rate limits and privacy logging.
- **Production:** HTTPS, managed secrets, request limits, monitoring and backup only for data actually stored.
- Commit only an .env.example with placeholders; keep real keys out of Git.
- Align store listing, permissions, privacy disclosures and behavior before release.

PART III — DESIGN.md

1. Design intent

Verity should feel like a calm, independent decision assistant—not an alarmist security scanner or sales funnel. Reduce cognitive load, surface consequential information first, and let users inspect evidence before trusting a conclusion.

> **Design principle: clarity over drama. Show what is known, what is inferred and what is missing.**

2. Experience principles

- **Evidence before verdict:** no dramatic badge without a concrete explanation and source.
- **Progressive disclosure:** summary first; details one click away.
- **Neutrality:** no manipulative urgency or affiliate-led ranking.
- **Explicit uncertainty:** say “Not found on this page,” not “does not exist.”
- **Actionable next step:** important findings offer a reasonable check.
- **Privacy visible:** explain what is analyzed and when data leaves the browser.

3. Brand and visual direction

Modern, quiet SaaS: warm off-white surfaces, deep ink text, restrained violet actions and soft semantic status colors. Avoid a dense finance-dashboard aesthetic. Red/green must never be the only signal.

Token	Value	Use
color.ink	#18212F	Primary text and headings
color.muted	#5C6778	Secondary text
color.primary	#6657D9	Primary action, links and focus accents
color.primary-soft	#F0EDFF	Selected surfaces/callouts
color.success-soft	#E8F7F0	Completed/positive state
color.warning-soft	#FFF4DB	Caution or missing terms
color.danger-soft	#FCE9E7	High-priority concern only
color.canvas	#F7F8FB	App background
color.border	#DDE3EA	Dividers and borders

Typography

- Use system UI / Inter-like sans-serif; headings semibold and body regular.
- Suggested sizes: title 20–22 px, section 14–16 px, body 13–14 px, support text 11–12 px.
- Use tabular numerals for prices where available.
- Avoid long all-caps labels and tiny legal text.

Spacing, shape and motion

- Spacing scale: 4, 8, 12, 16, 20, 24, 32 px.
- Cards radius 12–16 px; controls 8–10 px; subtle borders/shadows.
- Aim for 360–420 px extension width; avoid horizontal scrolling.
- Transitions 120–180 ms; respect reduced-motion preferences; no looping attention animation.

4. Information architecture

Area	Purpose	Priority
Check	Analyze current page; consent and progress.	Primary
Report	Cost snapshot, findings, unknowns and evidence.	Primary
Compare	Side-by-side comparison of another user-selected offer.	P1
Saved	Opt-in local reports and delete controls.	P1
Settings & privacy	Data handling, provider disclosure, clear saved data and feedback.	Required
Help / limitations	Supported pages, caveats and support route.	Required for public beta

5. Screen specifications

A — First-run / consent

- Brand + promise: “Understand the offer before you commit.”
- Plain disclosure: relevant current-page content is read and extracted information is sent to the service/AI provider to create a report.
- Primary action follows disclosure and any required consent.
- Secondary “How data is used” opens readable details.
- Do not hide provider processing or retention in a tooltip.

B — Ready to check

- Show page title and merchant domain so the analyzed page is obvious.
- Primary button: “Check this offer”.
- Unsupported page explains why and offers retry or manual-text path only if implemented safely.
- Helper: “Verity checks information available on this page; it may not see every checkout fee or term.”

C — Loading

- Compact progress/status: “Reading offer details,” “Checking cost structure,” “Preparing evidence.”
- Do not claim a stage is complete unless the backend reports it; an indeterminate loader is fine.
- Keep page identity visible; show timeout/retry if the request exceeds the limit.

D — Report (primary screen)

Order	Component	Behavior
1	Header	Merchant/page label, analyzed time, copy/delete menu where relevant.
2	Cost snapshot	Upfront, recurring, intro period and annualized estimate only when supported; show assumptions.
3	Important to know	Top 1–3 findings with severity text, explanation and evidence affordance.
4	Unknown / not found	Taxes, shipping, cancellation, refund or renewal terms not found in supplied content.
5	Evidence drawer	Exact snippet, source label and why it matters; no giant page dump.
6	Next steps	Practical checks, e.g. confirm renewal amount at checkout.
7	Actions	Compare, copy and opt-in save.

E — Compare

- Compare only normalizable fields; show “not available” for missing data.

- Align recurring cost, upfront cost, commitment, cancellation/refund terms and unknowns row-by-row.
- Avoid an overall score until a transparent scoring model is validated.
- Do not compare unlike currencies as equivalent.

F — Saved reports and settings

- Saved list shows merchant/title, date and price summary with delete control.
- Provide delete-one and clear-all; confirm destructive clear-all.
- Privacy settings explain data processing, retention and model provider plainly.
- Feedback form warns users not to include payment data or passwords.

6. Component inventory

Component	States	Rules
Primary button	Default, focus, disabled, loading	One primary action per screen; clear verb.
Finding card	Important, caution, informational	Text + icon + explanation; color is supplemental.
Evidence snippet	Collapsed, expanded, unavailable	Source text and relation to finding; never fabricate a quote.
Cost tile	Upfront, recurring, annualized, unknown	Currency and billing period always visible.
Unknown field	Not found, not applicable, extraction failed	Distinguish absence from technical failure.
Status banner	Success, caution, error, offline	Plain-language meaning and next action.
Consent disclosure	First use, provider/policy changes when required	Readable and explicit; link to full policy.
Tooltip/help	Contextual explanation	Critical caveats must not exist only in tooltips.

7. Content design and tone

Avoid	Prefer
“This site is a scam.”	“We found signals worth checking. This is not proof of fraud.”
“Hidden fee detected” when no fee is visible.	“Additional fees were not shown in the supplied page content.”
“Safe” / “100% legitimate.”	“No major concern was identified in the information checked; this is not a guarantee.”
“Best deal” without criteria.	“Lower recurring price, but cancellation terms were not found.”
“You will pay $X” when fees are unknown.	“Visible listed cost is $X; taxes and other checkout fees may apply.”

8. Accessibility and responsive behavior

- All actions keyboard reachable; logical tab order and visible focus ring.
- Aim for WCAG AA contrast for text and controls.
- Use text labels alongside icons/color; announce loading and errors accessibly.
- Support browser zoom and narrow windows without clipping essential content.
- Expandable evidence is keyboard accessible and exposes expanded/collapsed state.
- Respect reduced motion; never convey critical information only through animation.

9. Empty, error and edge states

State	Behavior
Unsupported page	“This page format isn’t supported yet.” Explain supported page types; do not imply risk.
No price found	“We couldn’t confirm a price from this page.” Retry; never invent an amount.
No terms found	Show unknowns and note terms may exist elsewhere, such as checkout or FAQ.
API timeout	Explain delay and offer retry; do not label a partial result complete.
Invalid model output	Return validated partial data if safe; otherwise explain failure.
Offline	Show connection state and retry; do not silently persist unsaved content.
Risk uncertainty	Downgrade to “needs checking” or omit when evidence is insufficient.

10. UX acceptance checklist

- New user understands purpose and data processing before first check.
- Report answers: cost, important terms, unknowns and evidence.
- Every amount includes currency and billing period.
- Every important finding traces to evidence.
- “Not found” is never presented as “does not exist.”
- Users can delete reports and understand third-party AI processing.
- Warnings are cautious, specific and actionable.
- Primary interactions work with keyboard and browser zoom.

APPENDICES

A. Example report (illustrative)

Offer snapshot
Introductory price: $5 for the first month
Then: $19.99 per month

Important to know
The page states that the price increases after the introductory month. [Evidence: introductory offer + renewal statement]

Not found in checked content
Taxes/checkout fees; refund policy.

Suggested next step
Confirm the renewal amount and cancellation steps before subscribing.

B. Suggested build sequence

Sprint	Work	Done when
0 — Validate	Interviews, corpus, schema and prototype.	Supported-page scope and evaluation plan documented.
1 — Extension	MV3 shell, active-tab flow, consent and basic extraction.	User can trigger mock analysis and inspect fields.
2 — Backend	FastAPI contract, normalizer and deterministic cost engine.	Unit/contract tests pass; unknowns remain unknown.
3 — Intelligence/UI	LLM adapter, evidence validation and report UI.	Every displayed finding has evidence or inference label.
4 — Hardening	Error states, save/delete, privacy logs and security review.	E2E flow passes; permissions and data flows reviewed.
Beta	Human evaluation, user interviews and fixes.	Quality gates met; limitations and support published.

C. Open decisions before implementation

- Will MVP support product pages, subscription pricing pages, or both? Recommended: both only if one extraction pipeline can handle them reliably.
- Which model provider and retention settings satisfy privacy and cost constraints?
- Will the backend retain any page data? Recommended default: transient processing only.
- Which currencies/locales are supported at launch?
- What snippet length and extraction limits balance quality and data minimization?
- Will accounts be omitted? Recommended: yes for MVP.
- What usage limits control model cost without breaking the core experience?
- What severity rubric and human review process define release readiness?

D. Launch gates

Gate	Must be true before public launch
Accuracy	Reviewed test set meets extraction/evidence goals; no critical unsupported fraud assertions.
Privacy	Actual data flows match disclosures; no unnecessary permissions or raw-content logs by default.
Security	Secrets protected; input limits/rate limits enabled; prompt-injection and URL handling reviewed.
UX	Consent, unsupported page, no-price, unknown, error and delete states tested.
Operations	Monitoring, provider-failure handling, support contact and rollback process documented.
Store readiness	Permissions, listing, privacy disclosure and behavior comply with Chrome Web Store requirements.

E. Glossary

Term	Meaning
Evidence	Source snippet/field extracted from the selected page that supports a finding.
Finding	User-facing observation about cost, terms or potential concern.
Unknown	Fact not established by the content available to the analyzer.
Deterministic calculation	Reproducible calculation implemented in code, not free-form model output.
MVP	Smallest version that tests the core value and trust assumptions.

Document owner: Product / Engineering • Version 0.1 • Review before implementation

This is a starting specification. Update it as user research, technical evaluation and accuracy testing produce evidence.